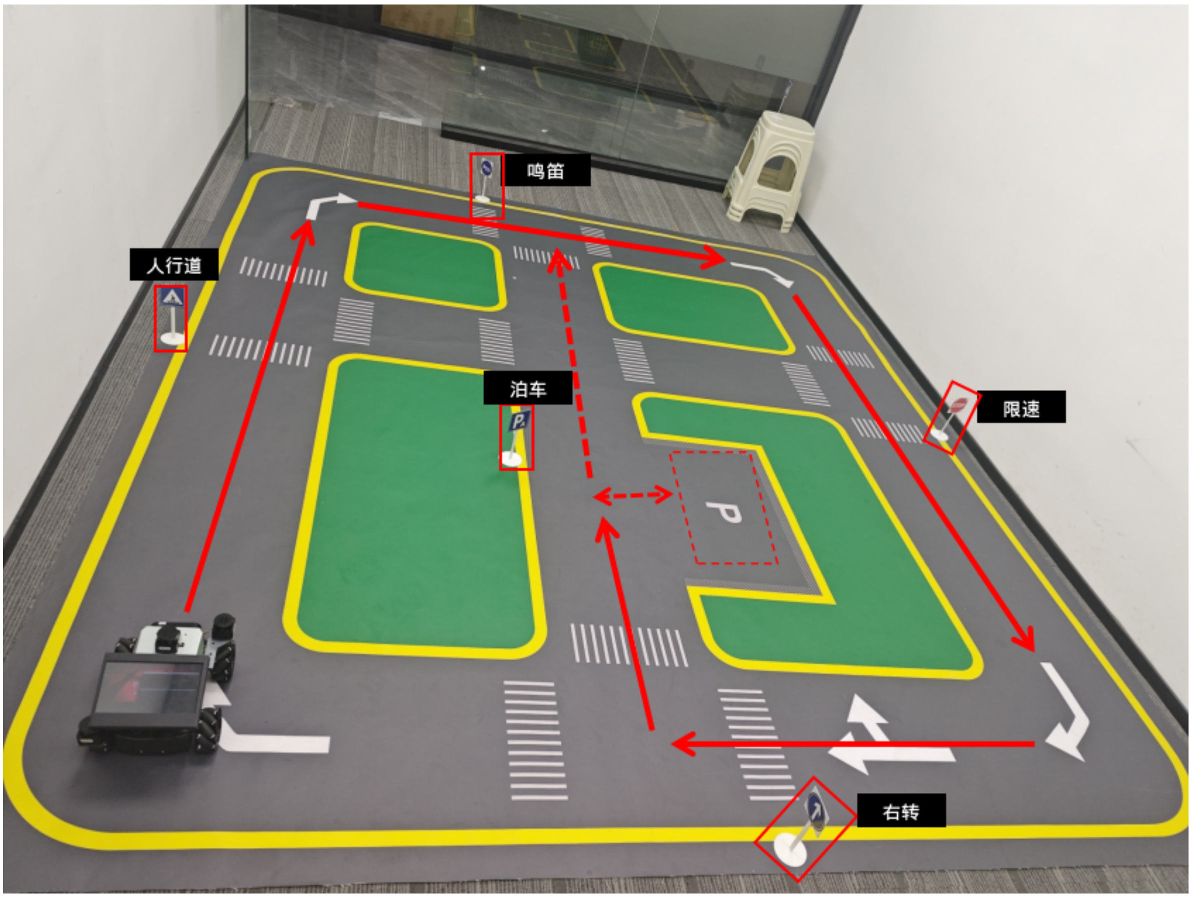
Comprehensive Autonomous Driving Applications

1. Course Content

- Start the autonomous driving program, combining lane-keeping autonomous driving and traffic sign recognition responses to complete the entire autonomous driving process in the sandbox map
- Site layout reference:

- Here is an example reference autonomous driving process as follows:
  - Lane-keeping driving ----------> Sidewalk ----------> Honking ----------> Speed limit ----------> Right turn ----------> Parking ----------> Exit parking ----------> Lane-keeping driving



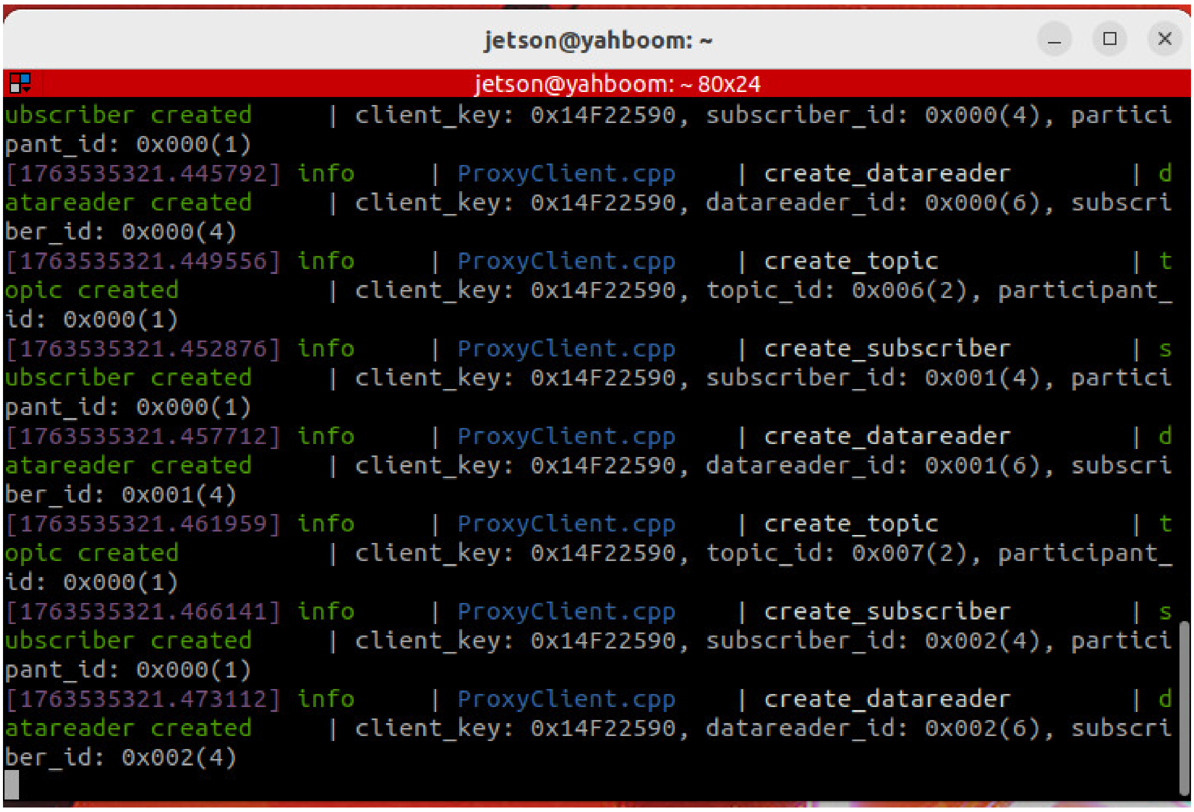
2. Start Example Program

2.1 Start Chassis Communication Agent

- Use the following command to start microROS agent to connect to the chassis

> [!WARNING]
>
> If the agent has already been started, there is no need to start it repeatedly

```
start_agent
```



2.2 Start Autonomous Driving Program

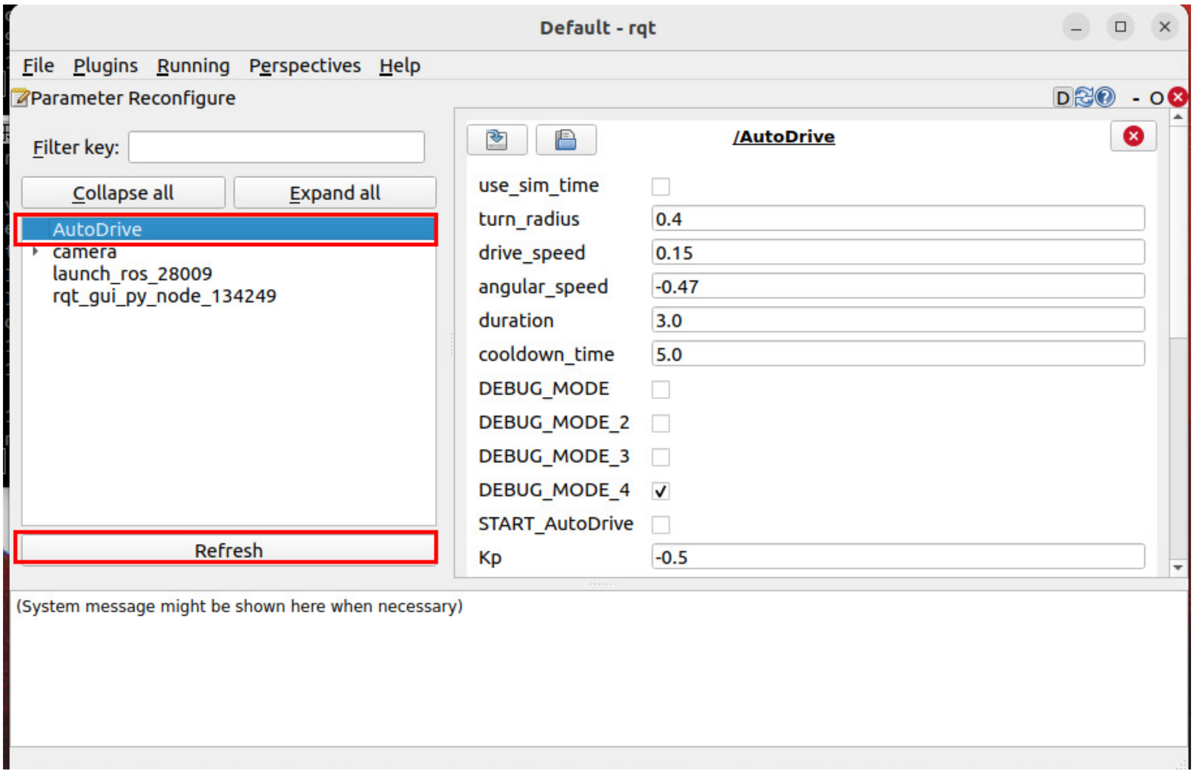
- Start the autonomous driving main program

```
ros2 launch m3_bringup auto_drive.launch.py
```

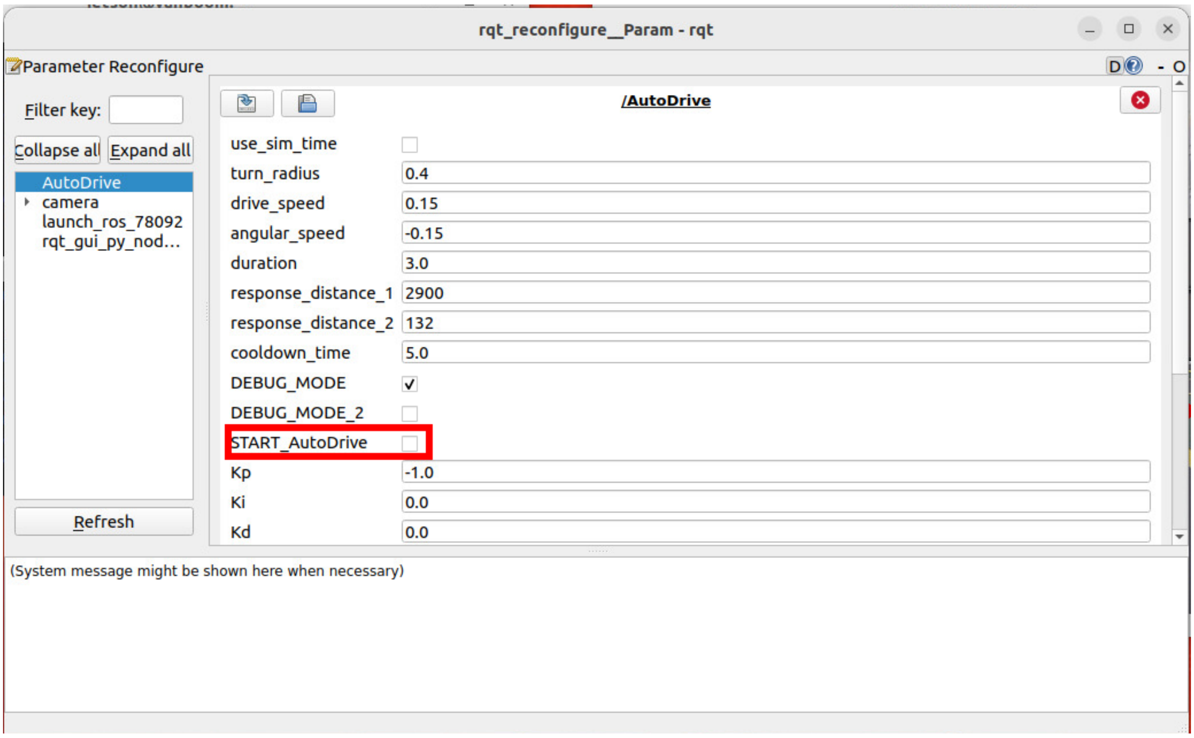
- Start dynamic parameter configuration panel

```
ros2 run rqt_reconfigure rqt_reconfigure
```

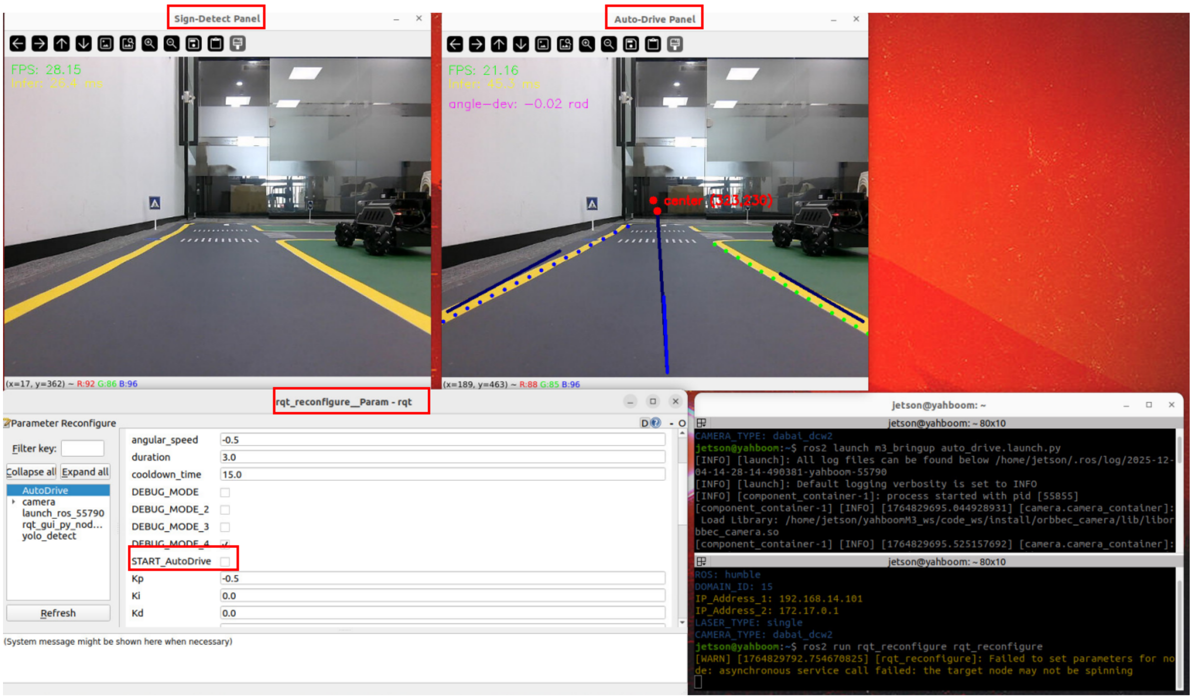
- Click AutoDrive to enter the parameter control panel. If there is no `AutoDrive` on the left side when you first open it, click `Refresh` below



- Click `START_AutoDrive` to start autonomous driving and lane-keeping functionality

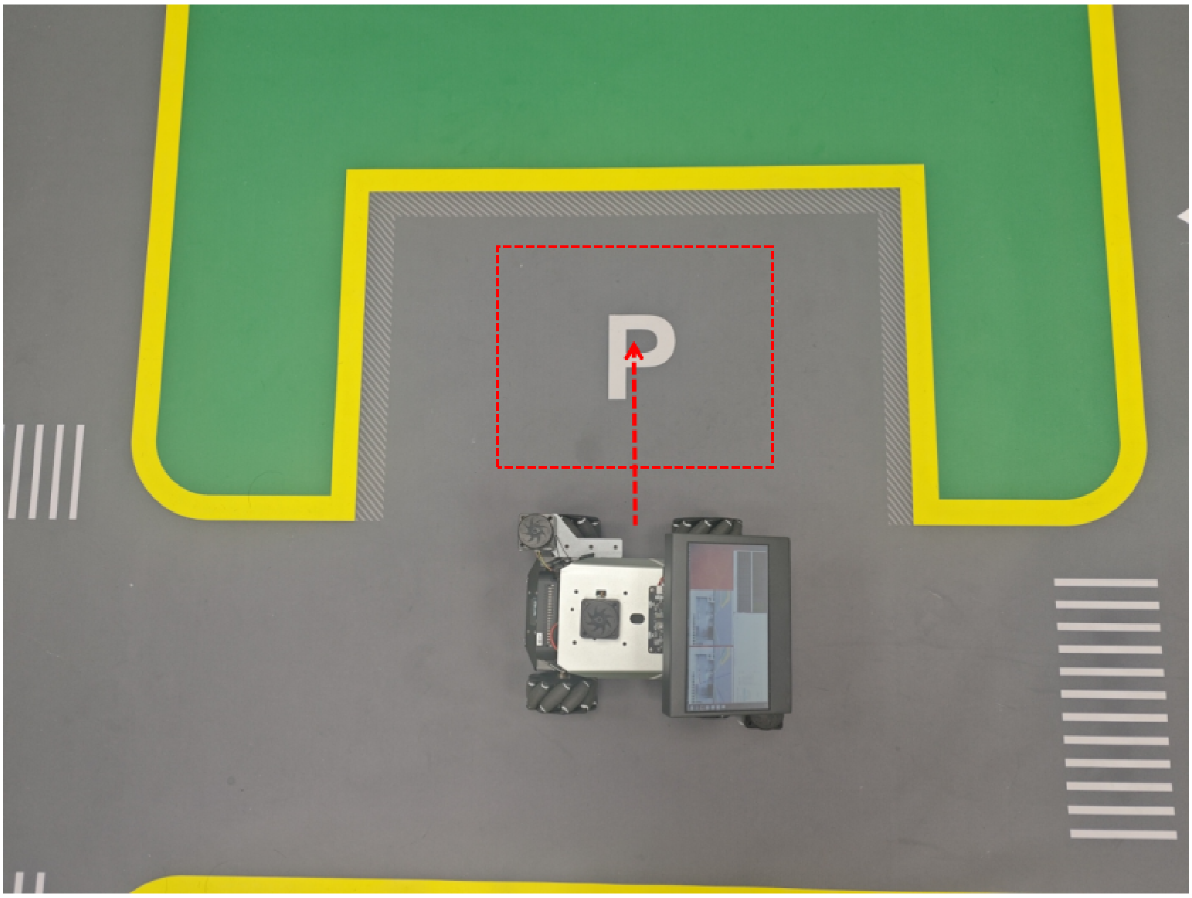


- After running the above commands, the startup interface is as follows, where
  - `Sign-Detect Panel` is the display window for YOLO traffic sign recognition
  - `Auto-Drive Panel` is the display window for autonomous driving lane line detection
  - `rqt_reconfigure` is a visual window used to dynamically adjust parameters and start autonomous driving switches
- Click the `START_AutoDrive` switch to start autonomous driving

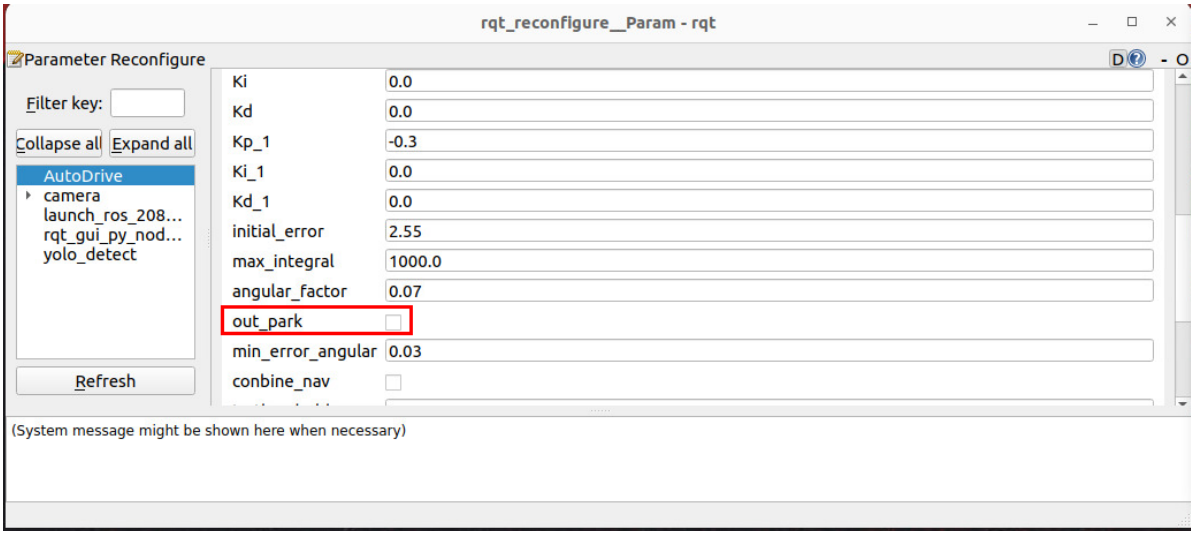


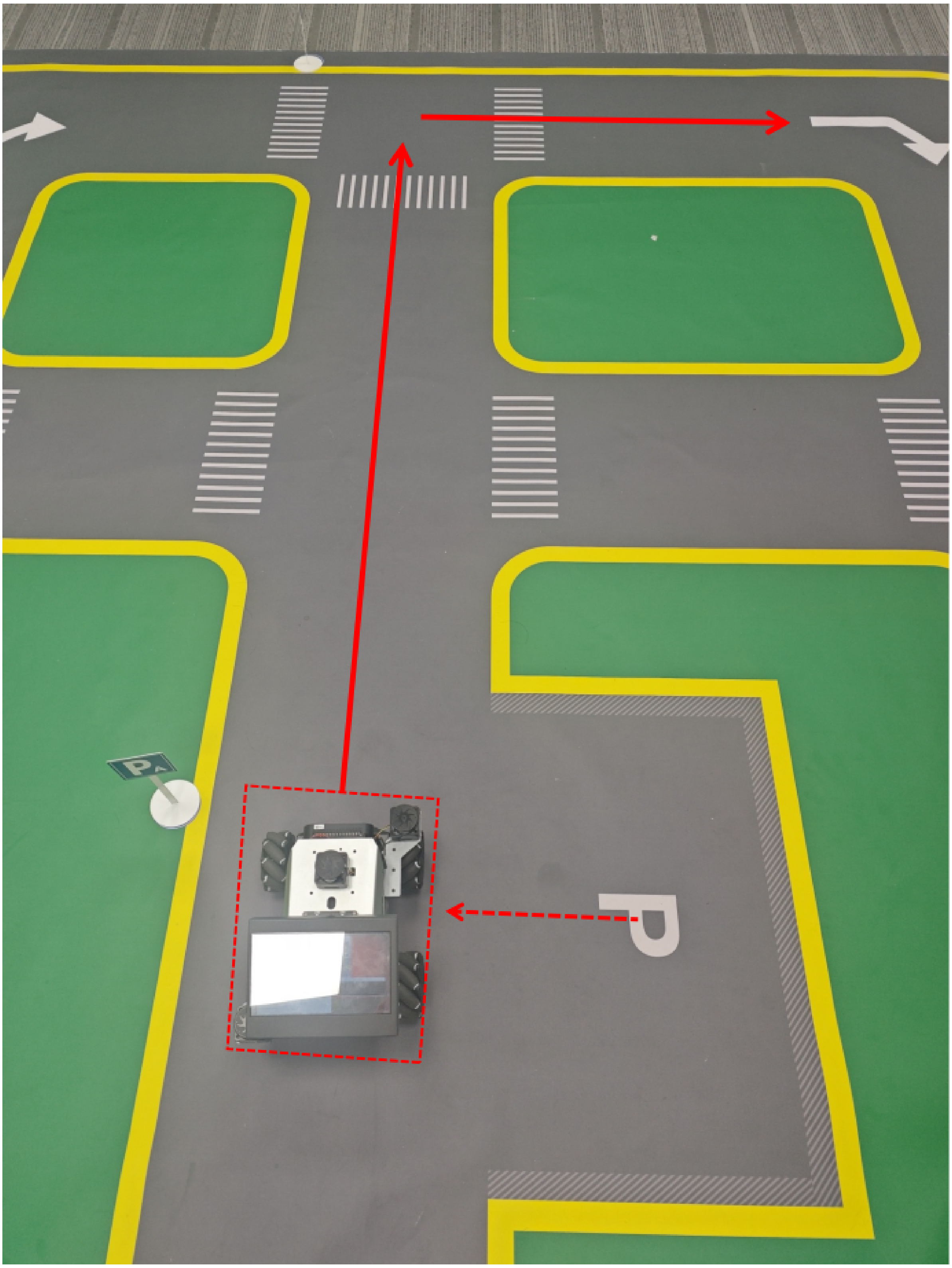
- Intersection turning
    - When a right turn road sign is detected, it will automatically move forward a certain distance to the three-way intersection in lane-keeping mode, then turn to the specified angle, and finally return to lane-keeping mode to continue moving forward.

- Autonomous parking
    - When road sign PA or PB is detected, trigger automatic autonomous parking, drive forward until aligned with the parking space, then translate into the garage.



- One-click exit parking
  - In the `rqt_reconfigure` panel parameters, click to check `out_park`, the vehicle will automatically translate out of the parking space, then return to lane-keeping autonomous driving mode to continue driving, and will turn right at the three-way intersection and continue driving.


rqt_reconfigure__Param - rqt
Parameter Reconfigure
Filter key:
Collapse all
Expand all
AutoDrive
camera
launch_ros_208...
rqt_gui_py_nod...
yolo_detect
Refresh
Ki
0.0
Kd
0.0
Kp_1
-0.3
Ki_1
0.0
Kd_1
0.0
initial_error
2.55
max_integral
1000.0
angular_factor
0.07
out_park
min_error_angular
0.03
conbine_nav
(System message might be shown here when necessary)



> [!TIP]
>
> - If the autonomous driving effect does not meet expectations, refer to this chapter's [12-Autonomous Driving Debugging Module] for parameter optimization.

3. Source Code Analysis

- The new actions that appear here are as follows:
- The basic principle is to control chassis movement through speed control, where the movement distance and rotation angle are adjusted through parameters.

```python
def turn_right(self):
    '''Turn'''
    self.get_logger().info('turn_right')
    drive_time=self.turn_distance/self.drive_speed
    time.sleep(drive_time)
    self.enable_auto_drive(False)
    self.__turn_by_angle(self.turn_angle)
    time.sleep(0.5)
    self.enable_auto_drive(True)
def parking_A(self):
    ''''Parking A'''
    self.get_logger().info('parking A')
    drive_time_1=self.park_distance_1/self.drive_speed
    drive_time_2=self.park_distance_2/0.2
    time.sleep(drive_time_1)
    self.enable_auto_drive(False)
    time.sleep(0.5)
    self.__set_current_speed(linear_y=-0.2)
    time.sleep(drive_time_2)
    self.__set_current_speed()
def parking_B(self):
    '''Parking B'''
    self.get_logger().info('parking B')
    drive_time_1=self.park_distance_1/self.drive_speed
    drive_time_2=self.park_distance_2/0.2
    time.sleep(drive_time_1)
    self.enable_auto_drive(False)
    time.sleep(0.5)
    self.__set_current_speed(linear_y=-0.2)
    time.sleep(drive_time_2)
    self.__set_current_speed()

def out_parking(self):
    '''Vehicle leaving the depot'''
    drive_time=self.park_distance_2/0.2
    self.__set_current_speed(linear_y=0.2)
    time.sleep(drive_time)
    self.enable_auto_drive(True)
```